CT071-3-3-DDAC - DESIGNING & DEVELOPING CLOUD APPLICATIONS

GROUP PROJECT

TASK #2 – CLOUD SERVICE INTEGRATION & MONITORING (CLO 2 – 20 Marks)

Hand in Week : **Week 11**
Submission Week : **Week 14**
Submission Date & Time :
Marking Type : Group

Submission Details **:**

- **System Submission:** Submit the complete system in a compressed ZIP file.
- **Final Report on Word Documentation**

Task #2 Development Notes:

Note: *If your team did not submit Task #1 and directly submitted Task #2, the marking will only depend on the criteria for Task #2. The Task #1 exercise will not receive any score.*

In ***Task #2***, your team is required to enhance the server-based system developed in ***Task #1*** by integrating serverless architecture components and refactoring specific parts into a microservices architecture. This includes incorporating additional AWS cloud services such as SQS, SNS, or S3 to further extend the system's capabilities. These enhancements should aim to improve scalability, flexibility, and maintainability. Upon completion, you must use AWS monitoring tools such as CloudWatch and/or X-Ray to observe system behaviour and evaluate the performance of the updated application.

Section 1 – Cloud Architecture Diagram

In this section 1, your team is required to develop a cloud architecture diagram that illustrates the updated system design, incorporating serverless components and microservices-based adjustments. Your team should be prepared to clearly explain the reasoning behind the expansion of the original architecture from ***Task #1***, and how the integration of serverless and microservices concepts contributes to enhancing system scalability, flexibility, and efficiency. The completed cloud architecture diagram must be included as part of the final report.

Section 2 – System Implementation (10 Marks)

In this section 2, your team is required to integrate a functional serverless architecture and microservices into the existing server-based system, as outlined in the cloud architecture diagram from Section 1. The integration should include the use of cloud services such as Amazon API Gateway and AWS Lambda, with each microservice being designed to function independently and communicate with other components.

Your microservices should interact with at least one (1) of the following AWS cloud services:

1. Amazon Simple Storage Service (S3)
2. Amazon Simple Notification Service (SNS)
3. Amazon Simple Queue Service (SQS)

Ensure that the serverless components and microservices are seamlessly integrated with the existing system as shown in the cloud architecture diagram from Section 1. The integration should enhance the system's modularity, scalability, and flexibility, enabling each microservice to scale independently while working together to provide the overall system functionality.

Section 3 – Performance Monitoring

In this section, your team must utilize cloud monitoring tools, such as ***AWS CloudWatch*** and/or ***AWS X-Ray***, to assess the performance of the serverless components, microservices, and the overall application developed in section 2. You should focus on monitoring key metrics such as system latency, error rates, and resource utilization to ensure optimal performance. The collected data will help evaluate the effectiveness of the serverless and microservices architecture integration and guide any necessary improvements to enhance scalability and reliability.

Section 4 – Final Report Structures (10 Marks)

Finally, document all application screenshots and performance analysis results in a Word document. The report should include the descriptions of the system functionality, the serverless and microservices integration, and any performance optimizations made during Task #2. Additionally, include relevant monitoring data from AWS CloudWatch and/or AWS X-Ray that illustrates the performance of the system post-implementation.

Submit the final Word document to Moodle by Week 14 using the following naming template:

G1_Task_2_Final Report.docx

Ensure that the document is organized, clearly written, and includes all required materials for assessment. This will be your final deliverable, summarizing both technical implementation and performance analysis.

Requirement of Word Document Content:

- Maximum pages: 40 pages
- Maximum Word Count: 4000 words
- Word document content should include the following:
  1. **Cover Page**
     - Selected Problem Statement Number & Topic
     - Project Title
     - Group Number
     - Team Member
  2. **Design & Implementation**
     - Cloud architecture diagram for Task #2
     - Discussion on the changes made to the cloud architecture between Task #1 and Task #2
     - Include relevant code screenshots demonstrating the implementation of serverless components (e.g., AWS Lambda functions, API Gateway configuration) and microservices (e.g., how each service is structured and interacts with the others).
     - Include screenshots that demonstrate the functionality of the system post-implementation.
  3. **Result and Discussion**
     - Attach results from AWS CloudWatch or AWS X-Ray, showcasing system performance, latency, error rates, and other key metrics.
     - Provide appropriate discussion on the performance analysis results
  4. **Reflection from each member.**
     - Provide a personal account from each team member on their experience during the development process for both Task #1 and Task #2.
     - Highlight challenges faced, lessons learned, and insights gained during the integration of serverless components, microservices, and cloud services.
     - Discuss how their individual contributions fit into the larger team effort and the overall system development.
  5. **New Workload Table Matrix**
     - Provide a detailed breakdown of each team member's responsibilities and contributions for both task #1 and task #2
  6. **References (APA style)**